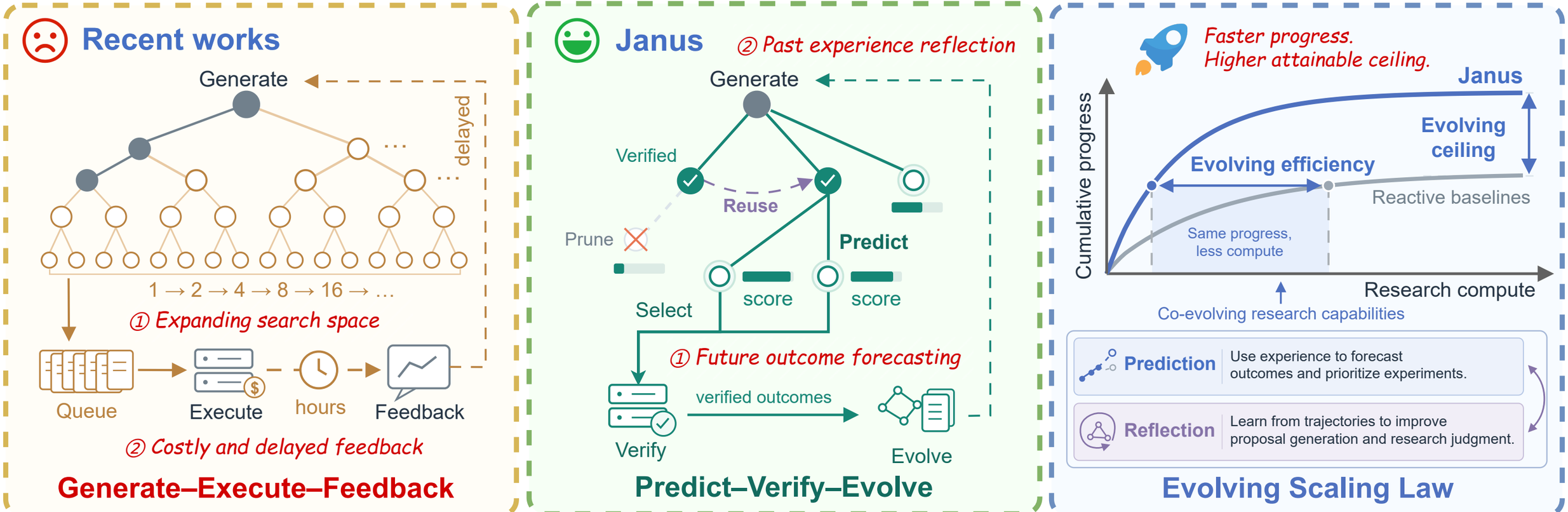

Recent works
Generate
delayed
1 → 2 → 4 → 8 → 16 → …
① Expanding search space
Queue
Execute
hours
Feedback
② Costly and delayed feedback
Generate–Execute–Feedback
Janus
② Past experience reflection
Generate
Verified
Reuse
Prune
Predict
Select
score
score
① Future outcome forecasting
verified outcomes
Verify
Evolve
Predict–Verify–Evolve
Faster progress.
Higher attainable ceiling.
Janus
Evolving ceiling
Evolving efficiency
Reactive baselines
Same progress, less compute
Cumulative progress
Research compute
Co-evolving research capabilities
Prediction
Use experience to forecast outcomes and prioritize experiments.
Reflection
Learn from trajectories to improve proposal generation and research judgment.
Evolving Scaling Law